

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed **AI-assisted documentation tooling** using **Python**, **LLM APIs**, and **RAG pipelines** to automate content classification, tagging, summarization, and structure validation across enterprise knowledge bases; integrated classification and link-validation services into CI/CD workflows, reducing manual content QA time by **40%**.
- Designed and implemented **AI-ready documentation transformations** for **DITA XML** content using **Python** and **data pipelines**, enriching structured content with metadata models, semantic tagging, and contextual embeddings to enable reliable LLM chunking, indexing, and reasoning; collaborated with documentation architects to define AI-readiness standards for **schema-driven** content delivery.
- Integrated **LLM-powered Q&A; and search assistants** into content platforms by building **REST APIs** and **microservices** using **Python (FastAPI)** and **TypeScript** that consume transformed DITA content; extended Azure DevOps CI/CD pipelines to orchestrate content enrichment without disrupting writer workflows, supporting **500K+** documentation pages.
- Built **content pipeline automation** using **Git-based workflows**, **Azure DevOps**, and **Python scripting** to validate DITA structure, enforce metadata consistency, and generate AI-consumption reports; reduced content-preparation bottlenecks by **55%** through automated pre-flight checks and schema validation.
- Collaborated across documentation leadership, platform engineers, and product teams to identify high-value **AI use cases** for documentation workflows; contributed technical guidance to the Documentation Community of Practice on structuring content for AI systems, including metadata standards, semantic enrichment, and prompt-engineering best practices for LLM integration.
- Managed **change-control workflows** and **content delivery pipelines** integrating **Oxygen/DITA authoring tools**, **OpenDITA frameworks**, and publishing systems; ensured backward compatibility and writer experience continuity while rolling out AI-assisted features, maintaining **100%** platform uptime across staging and production environments.

SKILLS

AI & Machine Learning – LLM API integration, RAG pipelines, prompt engineering, AI-assisted workflows, embeddings, semantic enrichment.

Programming Languages – Python, TypeScript/JavaScript, Java, SQL.

Structured Content & Documentation – DITA XML, schema-driven content, metadata modelling, semantic tagging, content transformation, Oxygen authoring tools, OpenDITA frameworks.

Data & Content Pipelines – Git-based workflows, Azure DevOps, CI/CD orchestration, data pipeline design, ETL, content validation automation.

Backend & API Development – REST APIs, FastAPI, microservices, API design for AI consumption, LLM integration services.

Content Platforms & Tools – DITA publishing systems, content management, Git workflows, Oxygen XML Editor, Azure DevOps pipelines, CI/CD integration.

Enterprise Systems & Collaboration – cross-functional stakeholder engagement, technical writing partnership, platform architecture, documentation operations.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India

01/2021

PROJECTS

AI-Ready DITA Documentation Enrichment Platform (HCL Technologies)

- Architected end-to-end **DITA XML transformation pipeline** using **Python** and **data engineering practices** to ingest unstructured documentation, apply **semantic tagging**, extract metadata, and generate **LLM-consumable embeddings**; processed **300K+** DITA topics monthly for AI inference.
- Built **FastAPI microservices** that expose transformed content via **REST APIs** for Q&A; assistants and search agents; implemented caching and indexing strategies reducing query latency from **2.5s to 480ms**, enabling real-time AI-powered documentation experiences.
- Integrated **content validation and quality automation** using **Python scripting**, **schema validation**, and **LLM-based link checking** into **Azure DevOps CI/CD pipelines**; detected and prevented **95%** of structural inconsistencies before publishing, reducing post-release documentation defects by **60%**.
- Designed **metadata enrichment framework** layering semantic context, versioning, audience tagging, and concept linking onto DITA structures; collaborated with documentation architects and product teams to define organizational standards, enabling consistent AI interpretation across **8 product lines**.
- Automated **documentation health reporting** with **Python dashboards** and **TypeScript visualization** components tracking content freshness, schema compliance, metadata completeness, and AI-readiness scores; provided visibility to leadership on documentation platform maturity across quarterly milestones.

LLM-Powered Documentation Assistance Toolkit (HCL Technologies)

- Developed **AI-assisted content generation and editing tools** using **LLM APIs** and **prompt engineering** to assist writers with summarization, code-example generation, cross-reference suggestions, and terminology standardization; deployed to **120+ technical writers**, reducing content creation time by **30%**.
- Built **classification and tagging services** in **Python** using **transfer learning** and **LLM-based zero-shot classification** to automatically tag DITA topics by product feature, audience level, and compliance domain; achieved **92% accuracy** on taxonomy assignment, eliminating manual tagging bottlenecks.
- Integrated AI tools into **Oxygen XML authoring environment** via **REST API plugins** and custom **TypeScript extensions**, ensuring writers could access AI assistance without leaving their workflow; measured adoption at **75%** of active users within **6 weeks** of launch.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)